

42/100

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct Reddy - 192372055

I M. Ganpikrishna (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H. G. 1

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

Expected concept nodes:

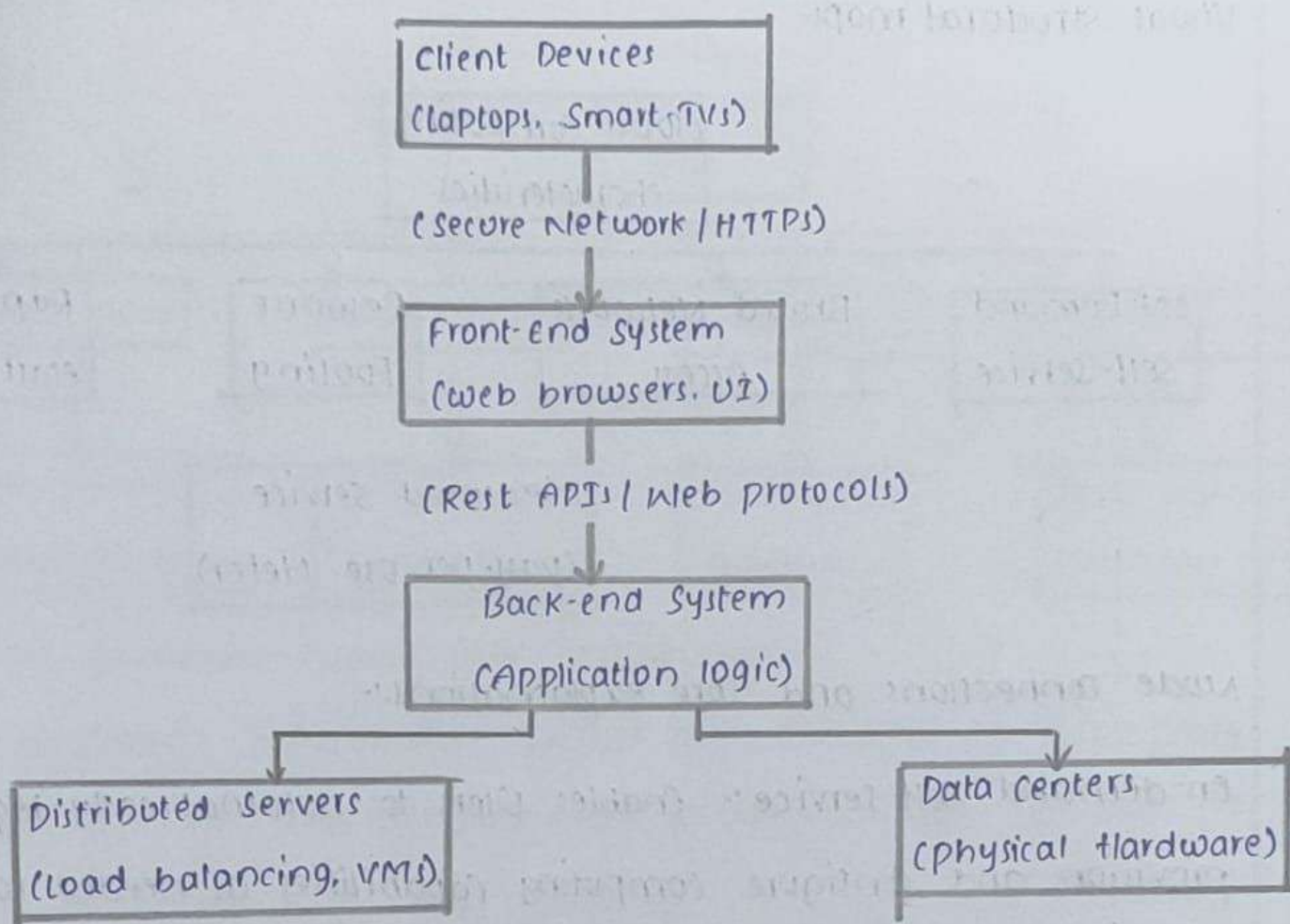
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

cloud Computing Architecture

Visual Structural Map:



Node connections and Core explanations:

Client devices: The local physical devices that users utilize to access cloud services over an internet connection.

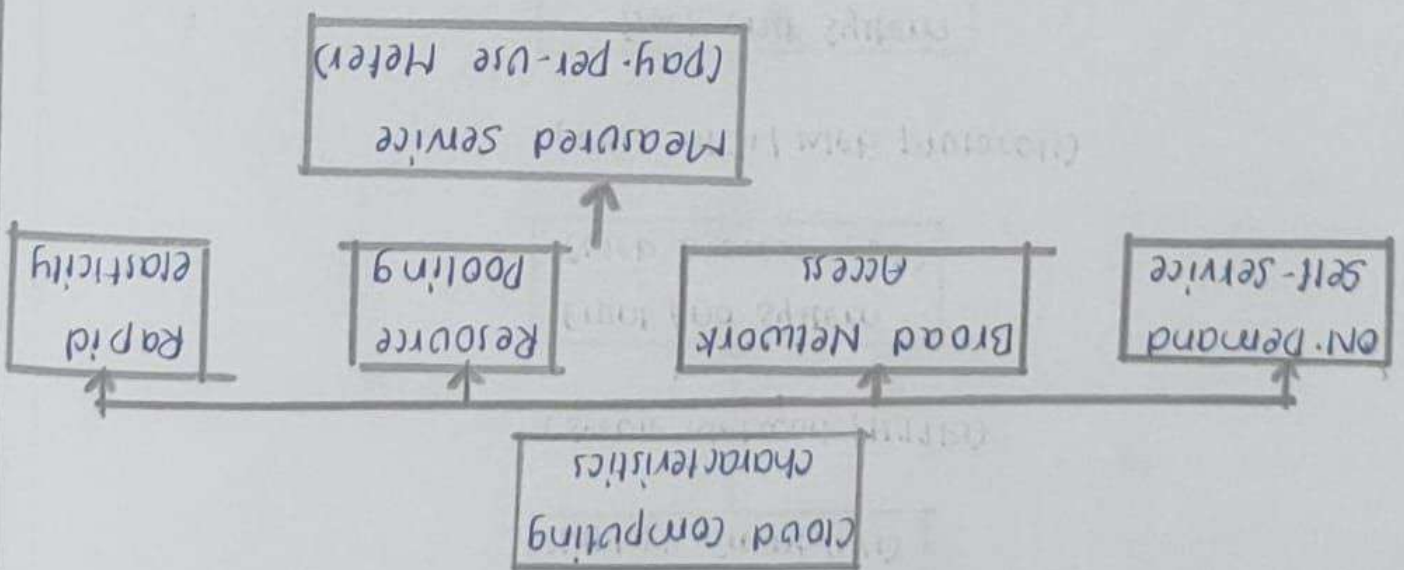
Front-end System: The client side interface consisting of web browsers, native mobile apps and graphical user interfaces. It collects user input and prevents data fetched from distant cloud servers.

Back-end System: The engine of the cloud architecture, comprising application processing runtimes that process business logic.

Data Centres: The physical facilities housing bare-metal server racks, central storage arrays, redundant power supplies and network routers that form the hardware base of the cloud.

2. Essential characteristics of cloud computing:

Visual structural map:



Node connections and core explanations:-

on-demand self-service:- enables users to automatically request, provision and configure computing capabilities as needed, without requiring human intervention for the service provider.

Broad network access ensures that cloud services are easily available over standard networks and can be accessed using heterogeneous thin or thick client platforms.

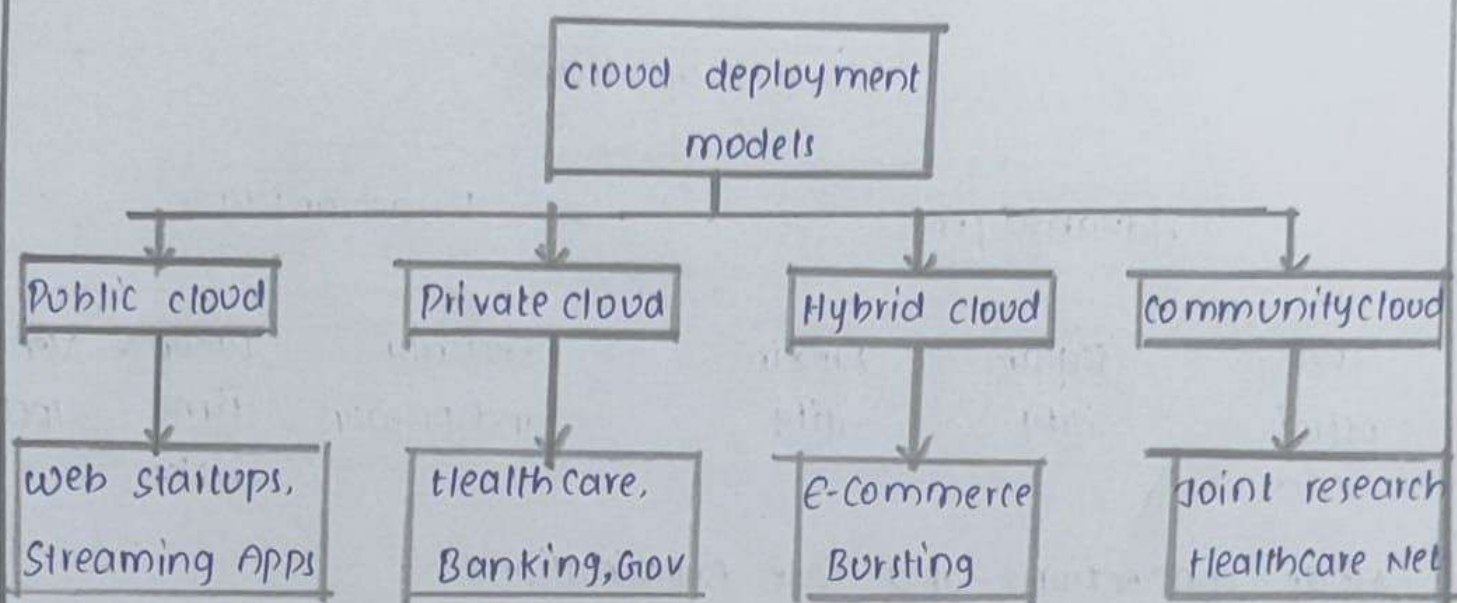
Resource pooling:- The providers' physical computing resources are pooled together to serve multiple customers using a multi-tenant model. Physical and virtual resources are dynamically assigned

according to consumer demand.

Rapid elasticity:- cloud capabilities can be elastically provisioned and released often automatically to scale rapidly outwards and inward in response to real-time workload fluctuations.

3. Deployment Models and Usage Scenarios.

Visual Structural Map:



Node connections and Core explanations.

Public cloud: Infrastructure owned and operated by third-party cloud service providers, shared among multiple organizations via multi-tenancy.

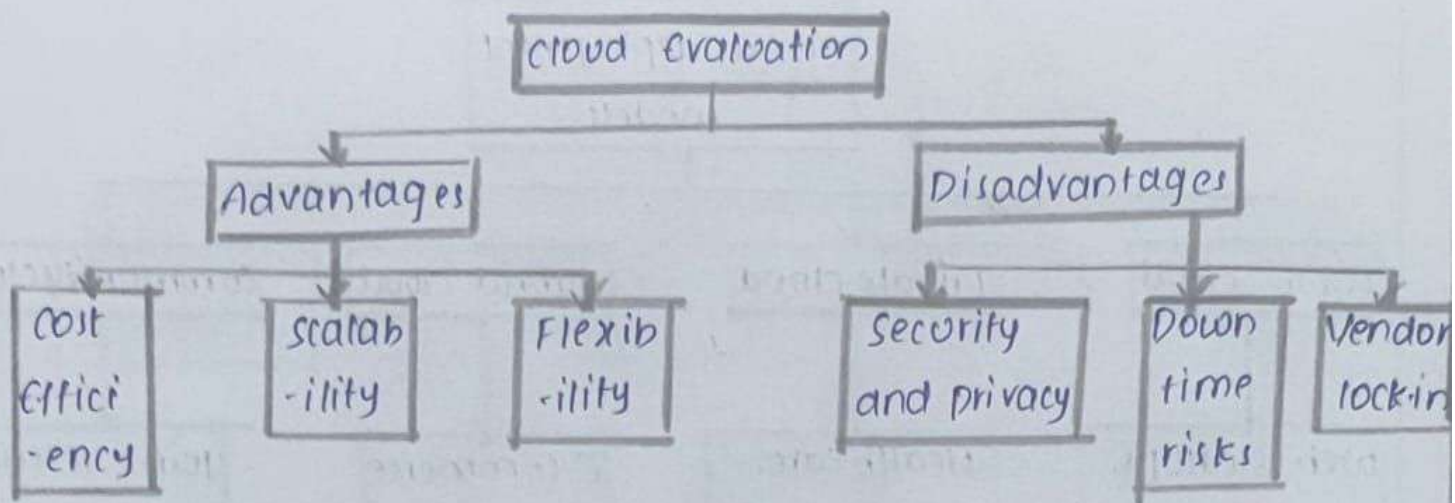
Private cloud: Cloud infrastructure operated solely for a single enterprise, hosted either on premises or managed off-site by a third party.

Hybrid cloud: An integrated infrastructure combining public and private cloud environments connected by standard technology that enables data.

Community cloud: Infrastructure shared by multiple organizations that have shared concerns, regulatory concerns, regulatory guidelines, or mission goals.

4. cloud benefits vs challenges

Visual Structural Map:



Node Connections and Core explanations:

Advantages:

Cost Efficiency: Replaces heavy upfront capital expenditure in physical hardware Opex pay-as-you-go model.

Scalability: Allows businesses to quickly expand or shrink resource capacity on demand to handle workload shifts effortlessly.

Flexibility: Enables employees to securely access critical software tools and shared data from any global location.

Limitations:

Security and privacy: Storing sensitive enterprise data on remote shared infrastructure introduces exposure risks, data breaches and regulatory compliance.

Downtime: Because cloud access depends on external network connectivity and provider infrastructure unexpected outages can temporarily stall businesses.